

IPCC Tier 2 Livestock GHG Uncertainty Calculator

User Guide

CIAT / CGIAR Alliance — ClimateActionNetZero

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Contents

1	Overview	3
1.1	What this tool does	3
1.2	Who this tool is for	3
1.3	Emission sources covered	3
2	Preparing your data	4
2.1	What you need before you start	4
2.2	AI-assisted data preparation (optional, free)	4
2.3	Complete parameter reference	5
2.4	Understanding the input structure	11
2.5	Defining sub-categories	11
2.6	Obtaining uncertainty estimates	11
2.7	Choosing a distribution	12
3	Loading your data (Tab 1 — Data Input)	13
3.1	Using the example datasets	13
3.2	Uploading your own data	13
4	Reviewing data quality (Tab 2 — QA/QC)	15
4.1	What the QA/QC checks do	15
4.2	What to do for each status	15
4.3	Documenting deviations	15
5	Reviewing uncertainty distributions (Tab 3 — Uncertainty)	16
5.1	The parameter table	16
5.2	Quick-set buttons	16
6	Correlations (Tab 4 — Correlations)	17
6.1	Why correlations matter	17
6.2	Quick guide — which option do I pick?	17
6.3	Automatic correlations from time-series data	17
6.4	Manual correlation matrix	18
6.5	Structural defaults (expert-elicited)	18

6.6	Coefficient (emission factor) correlations	19
7	Running the simulation (Tab 5 — Simulate)	21
7.1	Settings	21
7.2	Running the simulation	21
7.3	Convergence diagnostics	21
8	Understanding results (Tab 5 — Results)	23
8.1	Headline metrics	23
8.2	Uncertainty decomposition (AD vs EF)	23
8.3	By-system and by-category breakdown	23
9	Sensitivity analysis (Tab 6)	24
9.1	Reading the tornado chart	24
9.2	SRC vs PRCC	24
10	Generating the IPCC report (Tab 7 — IPCC Report)	25
10.1	IPCC Table 3.3	25
10.2	Downloads	25
11	IPCC reporting conventions	26
11.1	Where uncertainty results go in the national inventory report	26
11.2	What IPCC expert reviewers will check	26
12	Worked example — Country X (hypothetical dairy)	27

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1 Overview

1.1 What this tool does

National GHG inventories based on IPCC Tier 2 methods require not only a central estimate of emissions but also a quantification of their uncertainty. This tool automates that process for livestock (cattle) using Monte Carlo simulation:

1. You supply your country-specific Tier 2 activity data and emission-factor inputs, each with an uncertainty range.
2. The tool randomly samples from the uncertainty distributions you have defined (thousands of times) and propagates those samples through the full IPCC Tier 2 equation chain (Eq. 10.1–10.34).
3. It reports the resulting 95% confidence interval around your emission estimate, identifies which parameters drive the most uncertainty (sensitivity analysis), and formats the results for submission in the format of IPCC 2006 Vol. 1 Ch. 3 Table 3.3.

The approach follows IPCC Approach 2 (Monte Carlo) as described in IPCC 2006 Vol. 1 Ch. 3.

1.2 Who this tool is for

This tool is designed for national inventory practitioners who:

- Are already conducting or have recently completed a Tier 2 cattle GHG inventory (enteric fermentation, manure management, direct and indirect N₂O from pasture deposition)
- Have at least rough estimates of the uncertainty in their key activity data and emission factors
- Need to report the uncertainty range for inclusion in national inventory reports

The tool assumes basic familiarity with IPCC Tier 2 methodology. Users do not need prior experience with Monte Carlo methods, but a conceptual understanding of probability distributions is helpful.

1.3 Emission sources covered

IPCC Category	Source
3.A.1	Enteric fermentation — cattle (CH ₄)
3.A.2	Manure management — cattle (CH ₄)
3.A.2	Manure management — cattle (N ₂ O direct)
3.A.2	Manure management — cattle (N ₂ O indirect)
3.C.4	Direct N ₂ O from pasture/range/paddock deposition
3.C.5	Indirect N ₂ O from pasture/range/paddock deposition

2 Preparing your data

2.1 What you need before you start

Before opening the tool, assemble the following:

1. **Activity data:** For each animal sub-category, the population (N), body weight (BW), weight gain (WG), milk yield (for dairy), milk fat %, digestible energy (DE%), manure management system (MMS) fractions, and other Tier 2 inputs.
2. **Uncertainty ranges:** For each parameter, an estimate of its uncertainty — typically either:
 - A percentage uncertainty (e.g. $\pm 20\%$ representing the half-width of a 95% confidence interval), or
 - Explicit lower and upper bounds.
3. **Manure management fractions** (if running manure N_2O): The percentage of manure allocated to each management system (pasture, solid storage, liquid, etc.).

If you do not yet have uncertainty estimates, see Section 2.4 for guidance on obtaining them.

2.2 AI-assisted data preparation (optional, free)

If your raw inventory data lives in your own Excel or CSV files with column names that do not match the template, you can use the free **GMH Uncertainty Translator** on claude.ai to do the column-mapping and unit-conversion work for you. The Translator is a pre-built Claude Project loaded with the parameter catalogue, template schema, and ten worked mapping examples. It is free to use with a free claude.ai account and requires no installation.

Workflow. The chat with the Translator has two distinct inputs you provide: a **pre-flight questionnaire** that you *paste as text* into the chat, and your **raw data file(s)** that you *attach* when Claude prompts you. The questionnaire is not a separate upload — it is a one-page form whose contents you copy-paste into the chat box, so Claude knows about your inventory up front and can skip its warm-up questions.

1. Download the **pre-flight questionnaire** from the Resources tab and fill it out — country, year, IPCC version, your sub-categories, manure systems, data fields, uncertainty source (about 2 minutes).
2. Sign up for a free claude.ai account if you do not already have one, then click *Open the Translator on claude.ai* in the Resources tab.
3. In the Translator chat, **paste the filled questionnaire as your first message**. This is text you copy-paste, not a file you attach. Claude will summarise back what it understood and then ask you to upload your data.
4. **Attach your raw data file(s)** using the paperclip icon in the chat input box (**.xlsx**, **.csv**, or even a PDF / screenshot of a table).

5. Claude proposes a column mapping with confidence flags; confirm ambiguities (typically 2–5 short questions, e.g. *“Is your column **weight** body weight or mature weight?”*).
6. Download the produced `filled_template_for_app.xlsx` and upload it in Tab 1 (Data Input).

A 1-page **Getting Started** guide with screenshots is available from the Resources tab.

When to use. The Translator is most useful when (a) your data is in a non-standard shape, (b) your column names differ from the template (e.g. `cattle_pop`, `live_wt_lb`), (c) you have data in inconsistent units (lbs vs kg, fractions vs percentages), or (d) you want IPCC defaults filled in for the parameters you do not measure.

Quality control. The Translator does column-mapping, not analysis. Always (a) spot-check the key numbers (population, body weight, milk yield) by comparing with your original file, and (b) review the QA/QC tab after upload — any value the Translator filled from an IPCC default is flagged in amber there.

2.3 Complete parameter reference

The table below lists all parameters accepted by the tool. Required parameters depend on the emission sources you are running. Parameters marked *n/a* in the IPCC default column must be supplied from your national data. The tool will auto-fill missing optional parameters with IPCC defaults and flag them in Tab 1 (see Section 3.2).

Table 1: Complete parameter reference — all Tier 2 inputs accepted by the tool.

Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference	refer-
Activity data							
N	Number of animals in this sub-category	head	<i>n/a</i>	±10%	Normal	Eq. 10.1	
BW	Average live body weight (also TAM = Typical Animal Mass)	kg	275	±15%	Normal	Eq. 10.3 / 10.6	
MW	Mature (adult) body weight — used in NEg growth equation	kg	300	±10%	Normal	Eq. 10.6	
WG	Average daily weight gain — set 0 for non-growing adults	kg/day	0	±30%	PERT	Eq. 10.6	
Milk	Daily milk yield per <i>lactating</i> animal (not sub-category average; tool multiplies by pct_calving internally). Set 0 for non-dairy sub-categories.	kg/head/day	4.0	±20%	Normal	Eq. 10.8	
Fat	Fat content of milk	%	4.0	±10%	Normal	Eq. 10.8	
pct_calving	Fraction of females in this sub-category that give birth (calve) during the year, 0–1.	fraction	0.60	±20%	Beta	2019 Refinement Vol.4 Ch.10 §10.5	

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Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference	reference
MilkPR	Protein content of milk — feeds the milk-N term in Eq. 10.33 (N retention rates for cattle, where the 6.38 milk-protein-to-N conversion is defined). The 2019 Refinement gives the relation %MilkPR = $1.9 + 0.4 \times \% \text{Fat}$ (so 4% fat → 3.5% protein); illustrative values are tabulated in Annex 10A.1 / 10A.2 / 10A.3.	%	3.3	±10%	Normal	Eq. Annex 10A.1–10A.3	10.33;
Energy coefficients							
DE	Digestible energy as % of gross energy — typical range 45–75%	%	55.0	±15%	Normal	Eq. 10.14	
Cfi	Maintenance energy coefficient — depends on sex and lactation status	MJ/day/kg ^{0.75}	0.386	±30%	PERT	Table 10.4	
Ca	Activity coefficient for locomotion energy	dimensionless	0.17	±30%	Triangular	Table 10.5	
C	Growth coefficient for NEg equation — depends on sex (female 0.8, castrate 1.0, bull 1.2)	dimensionless	0.8	±30%	Triangular	Eq. 10.6	
Cp	Pregnancy coefficient — 0.10 for pregnant animals	dimensionless	0.10	±10%	Beta	Table 10.7	
hours	Daily working hours — set 0 when animals do no work (typical for dairy/beef cattle inventories)	hours/day	0	±20%	PERT	Eq. 10.11	

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Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
Tw	Mean daily temperature in winter (°C) — Cfi cold-climate adjustment. Leave blank or set 20 to disable.	°C	20	±25%	Normal	Eq. 10.2
Enteric fermentation						
Ym	Methane conversion factor (% of gross energy in feed converted to CH ₄)	%	6.5	±8%	PERT	Table 10.12
Manure management — CH₄						
Bo	Maximum CH ₄ producing capacity of manure	m ³ CH ₄ /kg VS	2019R: 0.13 (Other regions, low productivity cattle); 0.24 (intensive dairy NA/W.Europe); 2006 Africa = 0.10	±20%	PERT	Table 10.16(a) (Updated)
ASH	Ash content of manure (fraction of dry matter)	fraction	0.08	±25%	PERT	Eq. 10.24
UE	Urinary energy as fraction of gross energy	fraction	0.04	±25%	PERT	Eq. 10.24

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Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
CP	Crude protein content of diet — used to estimate nitrogen excretion	%	10.0	±15%	Normal	Eq. 10.32
Manure management — N₂O (managed storage)						
EF3_S	N ₂ O emission factor for managed manure storage (used in Eq. 10.25)	kg N ₂ O- N/kg N	0.005	asymmetric	PERT	Eq. 10.25; Table 10.21
Frac_GASMS	Fraction of managed manure N volatilised as NH ₃ /NO _x (used in Eq. 10.26)	N fraction	2006: 0.20; 2019R: 0.21 (Ch.10 Table 10.22 gives per-MMS values)	asymmetric	PERT	Eq. 10.26; 2019 Table 10.22
EF4	N ₂ O EF for atmospheric N deposition (used in Eq. 10.27)	kg N ₂ O- N/(kg NH ₃ -N + NO _x -N)	2006: 0.010; 2019R ag- gregated: 0.010; wet 0.014; dry 0.005	asymmetric	PERT	Eq. 10.27; Ch.11 Table 11.3
Frac_LeachMS	Fraction of managed N lost through leaching (used in Eq. 10.28)	N fraction	0.02	asymmetric	Lognormal	Eq. 10.28; 2019 Table 10.23
EF5	N ₂ O EF for N leaching/runoff (used in Eq. 10.29)	kg N ₂ O- N/kg N	2006: 0.0075; 2019R: 0.011	asymmetric	PERT	Eq. 10.29; Ch.11 Table 11.3

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Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
Manure management — N₂O (pasture/range/paddock)						
EF3_PRP	N ₂ O emission factor for kg N ₂ O-dung/urine deposited on pasture (used in Ch.11 Eq. 11.1)	N/kg N	2006: 0.02; 2019R: EF3 _{PRP, CPP} aggregated: 0.004; wet 0.006; dry 0.002	asymmetric	PERT	Ch.11 Eq. 11.1; Table 11.1
Frac_GasPRP	Fraction of N volatilised from dung/urine on pasture (used in Ch.11 Eq. 11.9)	fraction	0.21 (cattle)	asymmetric	PERT	Ch.11 Eq. 11.9; 2019 Table 11.3
Frac_LeachPRP	Fraction of N lost through leaching from pasture deposition (used in Ch.11 Eq. 11.10)	fraction	2006: 0.30; 2019R: 0.24 (wet) / 0 (dry)	asymmetric	PERT	Ch.11 Eq. 11.10; 2019 Table 11.3

Note on asymmetric uncertainty: Parameters flagged “asymmetric” use explicit lower/upper bounds from the IPCC tables rather than a symmetric $\pm\%$ formula. The template pre-fills these bounds automatically from the IPCC 2019 Refinement tables. *Note on pct_calving:* legacy column names `pct_lactating` and `pct_pregnant` are automatically renamed to `pct_calving` on upload.

2.4 Understanding the input structure

The tool uses an Excel template with a specific structure. Download the template from the **Data Input** tab (Tab 1) by clicking “Download Excel template”.

The template has the following key columns in the **Parameters** sheet:

Column	Description	Example
<code>cattle_type</code>	Broad cattle category	<code>dairy, non_dairy</code>
<code>aggregation_level</code>	Production system or region name	<code>Highland smallholder dairy</code>
<code>sub_category</code>	Animal sub-group	<code>cows, heifers, calves_female</code>
<code>parameter</code>	Parameter code (see Table 1)	<code>N, BW, Ym, DE, Bo</code>
<code>mean</code>	Central estimate (point value from your inventory)	<code>450 (BW in kg)</code>
<code>distribution</code>	Uncertainty distribution shape	<code>normal, pert, lognormal</code>
<code>uncertainty_pct</code>	Symmetric $\pm\%$ half-width of 95% CI	<code>20</code>
<code>lower</code>	Lower bound (alternative to <code>uncertainty_pct</code>)	<code>360</code>
<code>upper</code>	Upper bound	<code>540</code>
<code>param_type</code>	IPCC framing	<code>activity_data</code> or <code>coefficient</code>
<code>unit</code>	Unit of measurement	<code>kg, %, fraction</code>

Important: Each row is one parameter for one sub-category. A country with two production systems and five sub-categories each would have approximately 12–15 parameters $\times$ 10 sub-categories = 100–150 rows.

2.5 Defining sub-categories

Sub-categories should match your national inventory. Typical sub-categories include:

- Dairy: `cows` (lactating), `dry_cows`, `heifers`, `calves`, `bulls`
- Non-dairy beef: `breeding_cows`, `growing_males`, `heifers`, `calves`, `oxen`

Each sub-category needs its own rows for all relevant parameters. The values of `cattle_type`, `aggregation_level`, and `sub_category` together identify each group uniquely — they must be spelled consistently across all rows for the same group.

2.6 Obtaining uncertainty estimates

For activity data (N, BW, milk yields):

- **Survey-based estimates:** If populations are estimated from livestock surveys, the survey

sampling error provides a direct estimate of uncertainty. The 95% confidence interval from a survey translates directly to the **lower** and **upper** bounds (or to `uncertainty_pct` if symmetric).

- **Expert elicitation:** Ask the data provider to give a range within which they are 95% confident the true value falls.
- **IPCC defaults:** IPCC 2019 Refinement Chapter 11 provides uncertainty ranges for key emission factors. Use these as the starting point if no country-specific data is available.

For emission factors and coefficients (Ym, Bo, DE, MCF, etc.):

- **Literature review:** If you used a country-specific value from a published study, the study's reported confidence interval or standard deviation can be used.
- **IPCC default uncertainty ranges:** IPCC 2019 Refinement (Vol.4 Ch.10 / Ch.11) gives lower / upper bounds for several emission factors (e.g. EF_{3,PRP}, Ym, MCF). The template pre-fills these where IPCC publishes them; where IPCC does not publish a CI, the template falls back to ranges from Penman *et al.* (2000) and Monni *et al.* (2007) — these are clearly distinguished in the template footnotes.
- **Expert elicitation:** Ask the expert who provided the value to estimate the plausible range.

If you genuinely have no information about a parameter's uncertainty, a reasonable default for emission factors is ± 20 –40%. For population data with census-based estimates, ± 5 –15% is typical.

2.7 Choosing a distribution

Distribution	Shape	Best for
Normal	Symmetric bell curve	Activity data (N, BW) with symmetric measurement errors
PERT	Smooth bounded curve	Emission factors with a min, mode, and max
Triangular	Peaked triangle	Clear most-likely value with known hard limits
Uniform	Flat (equal probability)	Conservative fallback when only bounds are known
Lognormal	Right-skewed	Strictly positive values with right-skewed uncertainty (Bo, EFs)
Beta	Bounded [0, 1]	Fractions (pct_calving, Cp)

Recommended defaults:

- Use **Normal** for population (N) and body weight (BW) — typically measured with roughly symmetric errors
- Use **PERT** or **Lognormal** for emission factors (Ym, Bo, MCF, EF3–EF5, Frac parameters) — bounded below at zero with right-skewed uncertainty
- Use **Beta** for fractions bounded to [0, 1] (pct_calving, Cp)
- Use **Uniform** as a conservative fallback when you only know the plausible range

3 Loading your data (Tab 1 — Data Input)

3.1 Using the example datasets

Two built-in example datasets are included to let you explore the tool without uploading your own data:

- **Country X (hypothetical dairy):** A fictional East African highland dairy system — 12 parameters, dairy/cows, including a synthetic 5-year time series.
- **Country Y (hypothetical pastoral):** A fictional semi-arid pastoral beef system — 11 parameters, non-dairy/breeding cows, including a synthetic 5-year time series.

Select either from the “Country / Example Data” dropdown. The tool loads the data immediately and you can proceed directly to Tab 2.

3.2 Uploading your own data

Step 1 — Choose an IPCC Guidelines version. Before downloading the template, select the IPCC version your inventory follows:

- **IPCC 2006:** Uses the eight original manure management systems (MMS). Choose this if your national inventory is based on the 2006 Guidelines.
- **IPCC 2019 Refinement:** Adds four additional MMS types (anaerobic digester/biogas, aerobic treatment, solid storage covered/compacted, burned for fuel) and uses updated emission factors and N excretion equations from the 2019 Refinement. Choose this for inventories aligned with the 2019 Refinement.

The MMS dropdown in the downloaded template is automatically filtered to show only the manure systems valid for the version you selected.

Step 2 — Download and fill in the template.

1. Click “Download Excel template” to get the pre-formatted workbook
2. Fill in the **Parameters** sheet following the structure in Section 2.3
3. Fill in the **Manure_Management** sheet if you are running manure N₂O sources
4. (*Optional*) Fill in the **Parameter_TimeSeries** sheet if you want correlations computed automatically from historical data
5. Save the file as .xlsx and upload it using the file picker

Step 3 — Review upload notifications.

After a successful upload, a green notification confirms how many parameters were loaded. Any warnings appear as amber notifications.

Missing parameters and auto-fill: If your upload does not include all required parameters, the tool will:

- Still load and display all parameters that *were* supplied — you can review them immediately

in Tab 2 and Tab 3.

- Automatically fill in missing parameters using IPCC default values (see the "IPCC default" column in Table 1).
- Show a **yellow warning notification** listing exactly which parameters were auto-filled and with what values (e.g. "*3 parameters auto-filled from IPCC defaults: $Y_m = 6.5$, $B_o = 0.10$, $DE = 55.0$* ").
- Flag those parameters as **Missing** (purple/grey) in the QA/QC tab (Tab 2).

You should review whether the IPCC defaults are appropriate for your country context and, where possible, supply country-specific values before your final simulation run.

Common upload issues:

- `cattle_type` must be `dairy` or `non_dairy` (lowercase, no spaces)
- Distribution names must match exactly: `normal`, `pert`, `triangular`, `uniform`, `lognormal`
- If `uncertainty_pct` is left blank, both `lower` and `upper` must be filled
- `param_type` must be `activity_data` or `coefficient`

4 Reviewing data quality (Tab 2 — QA/QC)

4.1 What the QA/QC checks do

The QA/QC tab runs automated checks on your parameter values and flags any that are potentially problematic. Five statuses are possible:

Status	Colour	Meaning
Pass	Green	Value is within $\pm 50\%$ of the IPCC default (Vol.4 Ch.10 / Ch.11)
Warn	Amber	Value deviates 50–200% from the IPCC default — check and document
Fail	Red	Value deviates $>200\%$ from the IPCC default — likely a data entry error
Missing	Grey	Parameter not found in upload; auto-filled with the IPCC default (Vol.4 Ch.10 / Ch.11)

4.2 What to do for each status

Pass: No action needed. The value is within $\pm 50\%$ of the IPCC default for the parameter (Vol.4 Ch.10 / Ch.11).

Warn: The value deviates 50–200% from the IPCC default. This may be legitimate (country-specific data, distinct production system, or a specific climate regime — e.g. EF₄, EF₅ or EF_{3,PRP} differ between wet and dry climates) or it may be a data-entry issue. Possible actions:

- Confirm the value matches your country context
- If it is country-specific and well-justified, document the source in your inventory report
- If you intended to use a climate-disaggregated default, check IPCC Vol.4 Ch.11 Tables 11.1 / 11.3 for the wet/dry value
- If it is an error, correct it in your Excel file and re-upload

Fail: The value deviates more than 200% from the IPCC default. Most often this means a data-entry error (e.g. a live weight of 50,000 kg, a DE% of 150%). Correct the value before proceeding.

Missing: A required parameter was not found in the upload. The tool has auto-filled it with the IPCC default value from PARAM_CATALOGUE (the value and the IPCC equation/table source are shown in the Tab 1 notification). Review whether the default is appropriate for your country context and, where possible, replace it with country-specific data.

4.3 Documenting deviations

For any Warn or Info flags where you are deliberately using a country-specific value that deviates from the default, document the justification in your national inventory report. IPCC expert reviewers will look for this documentation.

5 Reviewing uncertainty distributions (Tab 3 — Uncertainty)

5.1 The parameter table

Tab 3 shows all parameters loaded from your data as an editable table. Each row shows:

- The parameter name, unit, and current mean value
- The distribution type currently assigned
- The uncertainty_pct or lower/upper bounds
- A visual preview of the distribution shape

You can edit values directly in the table. Clicking a cell opens an edit field.

5.2 Quick-set buttons

Two quick-set buttons are available for rapid exploration:

- **Set all activity data to Normal $\pm 15\%$:** Sets N, BW, and other activity data parameters to a Normal distribution with $\pm 15\%$ uncertainty. Click again to undo.
- **Set all coefficients to PERT:** Sets emission factors and other coefficients to PERT distribution. Click again to undo.

These are useful for a first exploratory run. Refine individual parameters before your final analysis.

6 Correlations (Tab 4 — Correlations)

6.1 Why correlations matter

If multiple parameters tend to move together (e.g. larger animals also tend to produce more milk), ignoring this relationship underestimates the uncertainty in the total emission estimate. For most Tier 2 analyses, ignoring correlations (the default “No correlations” setting) is a reasonable simplifying assumption. IPCC Approach 2 guidance recommends including known correlations where they exist.

6.2 Quick guide — which option do I pick?

- Do you have ≥ 5 years of national time-series data in your upload? → **From template (auto, time-series)**.
- No time series, but you want to capture well-known biological linkages (BW $\leftrightarrow$ MW, Milk $\leftrightarrow$ Fat, DE $\leftrightarrow$ Ym, ...)? → **Structural defaults (expert-elicited)**.
- You have your own correlation matrix from a study or expert elicitation? → **Advanced — manual entry**.
- None of the above? → leave **No correlations** (the default; conservative and defensible).

6.3 Automatic correlations from time-series data

If you included a Parameter_TimeSeries sheet in your upload, the tool can compute historical correlations automatically:

1. On Tab 4, select **From template (auto, time-series)** under Activity Data Correlations.
2. The tool computes the **Spearman** rank correlation matrix of the historical parameter data, applies the chosen detrending treatment, and samples using **Iman–Conover restricted pairing** — the method explicitly cited by IPCC V1 Ch3 §3.2.3.2 (p. 26). Iman–Conover is distribution-free and reproduces the target rank correlation exactly, regardless of the marginal shapes.
3. Inspect the heatmap that appears below the controls to confirm correlations are plausible.

Treatment of trends. Most national livestock series share long-run growth (population up, milk yield up). The Pearson/Spearman correlation of two upward-trending series is mechanically high, but that trend co-movement is *not* the year-to-year joint uncertainty Approach 2 is meant to propagate. The tool offers three options:

- **First differences (recommended, default)** — year-on-year changes. Isolates the genuine year-to-year co-movement.
- **Linear detrend** — subtracts a fitted linear trend. Use when the trend is approximately linear and you want to keep more of the series structure.
- **Raw series** — legacy behaviour. Use only if the input is already stationary (e.g. you uploaded

already-differenced data).

Subgroup scope. You can restrict the matrix to a subset of parameters:

- **All parameters** (default) — correlations apply across the full parameter set.
- **Population-related only** (N, BW, MW, WG) — when only the population/animal-size variables are jointly uncertain.
- **Intake / feed-quality only** (DE, CP, Ym, Cfi, Ca, ...) — when the diet-related parameters share an estimation source but population data are independent.

Parameters outside the chosen subgroup are sampled independently of everything else.

6.4 Manual correlation matrix

If you prefer to specify correlations manually:

1. Select **Advanced — manual entry** under Activity Data Correlations.
2. Upload a square, symmetric CSV matrix of correlation coefficients (-1 to $+1$, diagonal = 1). The matrix is interpreted as a Gaussian-scale correlation (the value you enter for cell (i, j) is the correlation of the underlying normal copula).
3. The tool verifies positive-definiteness and projects onto the nearest positive-definite correlation matrix if needed; you are warned if any entry shifts by more than 0.01.

6.5 Structural defaults (expert-elicited)

A built-in preset of structural defaults is available. It applies the following well-documented pairs, drawn from the IPCC equations and the livestock literature. *These values are expert-elicited; the IPCC publishes no numerical correlation coefficients for these pairs.* Hover any non-zero cell in the heatmap for the per-pair source citation.

Parameter A	Parameter B	ρ	Source
BW	MW	0.85	Growth-curve asymptote (Brody 1945); structural link via IPCC Eq 10.3 / 10.6. Assumes BW and MW come from the same census; ~ 0.5 is more typical if from independent surveys.
BW	WG	0.40	Structural link via IPCC Eq 10.6 joint estimation (NRC 2001 energy-balance studies, published range 0.30–0.50).
Milk	Fat	−0.30	Milk dilution effect (Wilmink 1987; VandeHaar 1998); within-herd literature −0.20 to −0.40.
Milk	pct_calving	0.20	Herd-management intensity linkage (expert judgement; conservative).
DE	CP	0.50	Forage-quality co-variation across feed types (NRC 2001; INRA; Feedipedia).
DE	Ym	−0.50	<i>Strengthened from −0.40 in May 2026 audit re-review.</i> IPCC 2019 Refinement Eq 10.21 (Ym decreases with DE) + meta-analysis (Niu et al. 2018 GCB; Hristov et al. 2013).
Cfi	Ca	0.30	<i>Lowered from 0.60 in May 2026 audit re-review.</i> Weak linkage as both are energy-equation constants. The previous 0.60 invoked an estimation-covariance argument that does not apply: Cfi and Ca are independent inputs in IPCC Eq 10.3 / 10.4, not jointly estimated.

All other pairs are set to zero. The previous $N \leftrightarrow BW = 0.30$ cross-block pair was removed in the May 2026 statistical audit (no defensible cross-country source).

6.6 Coefficient (emission factor) correlations

The right-hand card on Tab~4 controls correlations *between* the per-head IPCC coefficients (Cfi, Ca, Ym, Bo, MCF, EF3, EF4, EF5, ...). Two modes are available:

- **No EF correlations** (default) — coefficients are sampled independently. This is the IPCC Approach 2 default and is appropriate when each emission factor comes from its own study.
- **Block-structured EF correlation** (recommended when you do want non-zero EF correlations) — the IPCC coefficients are grouped into three blocks, each with its own within-block ρ

slider in $[0, 0.5]$:

- *Energy-equation coefficients* (Cfi, Ca, C, Cp, Ym) — rumen-fermentation studies (IPCC Eq 10.3 / 10.16 / 10.21 family).
- *Manure-CH₄ coefficients* (Bo, MCF, ASH) — BMP assays and lagoon-temperature studies.
- *Manure-N coefficients* (EF3, EF4, EF5, Frac_GASMS, Frac_LEACH, UE) — NH₃ / N₂O volatilisation studies.

Cross-block correlation is zero — these three literatures are independent. $\rho = 0.3$ is a moderate assumption capturing shared measurement bias within a block.

Note: a single- ρ “Uniform EF correlation” option was removed in the May 2026 audit. Applying one ρ to all coefficients implies that rumen-fermentation, BMP-lagoon and NH₃ volatilisation studies share a common bias, which is statistically indefensible.

7 Running the simulation (Tab 5 — Simulate)

7.1 Settings

Before clicking Run, configure:

Number of iterations: The number of Monte Carlo samples.

- **1,000:** Quick test run to check the model runs correctly. Do not use for reporting.
- **10,000:** Minimum for final reporting purposes. Tail percentiles (2.5th/97.5th) are reliable.
- **25,000–30,000:** Recommended for final reporting, especially when correlations are enabled or you have many sub-categories.

GWP version: Choose between AR5 (CH₄ GWP = 28, N₂O = 265) and AR6 (CH₄ GWP = 27.9, N₂O = 273). Select the version required by your national inventory reporting guidelines.

Random seed: Fixing the seed ensures results are exactly reproducible. Change the seed to verify convergence: if the 95% CI changes substantially between runs, increase the number of iterations.

Analysis mode: Single-year (most common) or Trend (multi-year time series).

Emission sources: Tick all sources applicable to your inventory. Unticked sources contribute zero emissions to the total.

7.2 Running the simulation

Click **Run Monte Carlo Simulation**. A progress bar shows the simulation steps. For 10,000 iterations and a single production system, typical run time is 15–60 seconds.

When complete, results appear in the lower section of the same tab.

If something is missing or not working: All settings (iterations, GWP, seed, emission sources) remain active in the upper section of Tab 5 — simply adjust any setting and click **Run Monte Carlo Simulation** again. The previous results are replaced automatically. If you need to change your input data (parameters, distributions, or correlations), go back to the relevant tab (Tab 1–4), make the change, then return to Tab 5 and re-run.

7.3 Convergence diagnostics

Click “Run Diagnostic” (magnifying glass button) after a simulation to assess whether enough iterations were run for stable results:

- **MCSE%** (Monte Carlo Standard Error %): Pass < 0.5%, Warn 0.5–1%, Fail ≥ 1%
- **Mean drift %:** How much the mean from the first half of iterations differs from the second half. Pass < 2%
- **CI drift %:** How much the 95% CI bounds shift between the two halves. Pass < 5%

- **Convergence trace:** Running mean and 95% CI plotted against iteration count — should stabilise before the end of the run

Note: this tool uses **independent Monte Carlo**, not MCMC. Each iteration is drawn independently — there are no chains, no warmup, and no burn-in.

8 Understanding results (Tab 5 — Results)

8.1 Headline metrics

After a successful run, five value boxes show:

- **Total CO₂eq (t)**: The mean of the simulated total emission estimate, in tonnes CO₂-equivalent
- **Enteric CH₄ (t)**: Mean enteric fermentation contribution
- **Manure CH₄ (t)**: Mean manure management CH₄ contribution
- **Pasture N₂O (t)**: Mean combined direct and indirect N₂O from pasture deposition
- **Total CV (%)**: Coefficient of variation = $SD / \text{mean} \times 100$. Note: the IPCC reporting metric is the % uncertainty (MoE%) shown in the IPCC Report tab.

8.2 Uncertainty decomposition (AD vs EF)

If “Run uncertainty decomposition (AD/EF/Combined)” was checked, the results include a breakdown:

- **AD uncertainty**: Driven by activity data only (N, BW, Milk, WG) with all emission factors fixed at their means
- **EF uncertainty**: Driven by emission factors and coefficients only with activity data fixed at means
- **Combined uncertainty**: Both sources varying together

If AD uncertainty dominates, improving livestock population surveys would most effectively reduce total uncertainty. If EF uncertainty dominates, investing in country-specific EF measurement studies would be more impactful.

8.3 By-system and by-category breakdown

- **By-system breakdown**: Mean, 95% CI, MoE%, and CV% for each production system and emission source
- **By reporting category (IPCC Table 3.3 layout)**: Same information grouped by IPCC source category (3.A.1, 3.A.2, 3.C.4, 3.C.5)

9 Sensitivity analysis (Tab 6)

9.1 Reading the tornado chart

The tornado chart ranks parameters by their contribution to total uncertainty. The longest bars indicate the parameters that, if better constrained, would most reduce the 95% CI.

Each bar is labelled with:

- **Cattle type and sub-category:** e.g. dairy | cows
- **Parameter name:** e.g. Y_m (methane conversion factor), N (population), BW (body weight)

Bars are coloured by reducibility:

- **Green:** Uncertainty can be reduced with better local data (surveys or measurement studies)
- **Grey:** Uncertainty is largely irreducible with national-level data (e.g. fundamental biochemical constants)

9.2 SRC vs PRCC

Two sensitivity methods are computed:

- **SRC** (Standardised Regression Coefficient): Linear sensitivity. Reliable when relationships are approximately linear.
- **PRCC** (Partial Rank Correlation Coefficient): Rank-based, captures non-linear relationships. More robust but less interpretable.

Both are shown; they typically agree on the top-ranked parameters. If they disagree significantly, the emission pathway may be strongly non-linear.

10 Generating the IPCC report (Tab 7 — IPCC Report)

10.1 IPCC Table 3.3

The IPCC Report tab shows your results formatted as IPCC 2006 Vol. 1 Ch. 3 Table 3.3. The three uncertainty columns show:

- **AD uncertainty (%)**: % uncertainty (MoE%) from activity data alone
- **EF uncertainty (%)**: % uncertainty (MoE%) from emission factors alone
- **Combined uncertainty (%)**: Overall % uncertainty (MoE%)

Definition of % uncertainty: Following IPCC Table 3.3 conventions, all values are expressed as the half-width of the 95% confidence interval divided by the mean, multiplied by 100.

Example: a Combined uncertainty of 35% means the 95% CI is approximately $\pm 35\%$ of the mean emission estimate.

10.2 Downloads

Three download formats are available:

- **Excel report (.xlsx)**: Full workbook with run settings, IPCC summary table, uncertainty metrics, and sensitivity results (SRC and PRCC)
- **CSV report**: The IPCC Table 3.3 formatted table only, for easy import into inventory reporting software
- **Word report**: Narrative summary suitable for inclusion in the uncertainty section of your national inventory report

11 IPCC reporting conventions

11.1 Where uncertainty results go in the national inventory report

IPCC 2006 Vol. 1 Ch. 3 §3.3 describes the standard approach for Approach 2 (Monte Carlo):

1. Report the 95% confidence interval for each major source category and for the national total
2. Present the results in Table 3.3 format (included in the IPCC Report tab downloads)
3. Describe the Monte Carlo approach used (number of iterations, distributions chosen, correlation assumptions)
4. Document any parameters where country-specific values were used instead of IPCC defaults, including the justification

The methodology PDF (available on the Resources tab) provides a complete technical description of the equations, sampling approach, and convergence assessment that can be referenced in your national inventory report.

11.2 What IPCC expert reviewers will check

Based on experience with IPCC expert reviews, reviewers typically focus on:

- Whether the uncertainty metric reported is the correct one (% uncertainty = MoE%, not CV%)
- Whether the IPCC Table 3.3 format and column conventions are followed
- Whether Approach 2 (Monte Carlo) is described sufficiently to be reproducible
- Whether the number of iterations is sufficient for stable results
- The basis for any country-specific emission factor values that deviate substantially from IPCC defaults

12 Worked example — Country X (hypothetical dairy)

To illustrate the workflow, the following example uses the Country X built-in dataset.

Step 1 — Load data: Select “Country X (hypothetical dairy)” from the dropdown on Tab 1. The data loads automatically (12 parameters, dairy smallholder system).

Step 2 — QA/QC: Go to Tab 2. All parameters should pass; any Warn flags would compare against the IPCC 2019 Refinement defaults (Vol.4 Ch.10 / Ch.11), and a deliberate country-specific value can be documented in the inventory report rather than corrected.

Step 3 — Uncertainty: Go to Tab 3. Note that activity data (N, BW, Milk, WG, Fat) use Normal distributions and coefficients (Ym, DE, Bo, etc.) use PERT or lognormal distributions.

Step 4 — Correlations: Go to Tab 4. Select “From template (auto, time-series)” — the example dataset includes a 5-year time series from which a Spearman rank correlation matrix is computed automatically. Leave “Treatment of trends” at “First differences” (the recommended default) and “Apply correlations within” at “All parameters”. The matrix is sampled using Iman–Conover restricted pairing. Leave the right-hand card at “No EF correlations” for the default run.

Step 5 — Simulate: Go to Tab 5. Set iterations to 10,000, GWP to AR6, seed to 42. Click “Run Monte Carlo Simulation”.

Step 6 — Results: After about 30 seconds, results appear. Note the 95% CI and MoE%. Click “Run Diagnostic” to confirm convergence.

Step 7 — Sensitivity: Go to Tab 6. The tornado chart shows which parameters drive most uncertainty. In this example, Ym and BW for cows are typically the dominant contributors.

Step 8 — Report: Go to Tab 7. Download the Excel report to see the full IPCC Table 3.3 output.

For questions, contact Lolita Muller (Alliance Bioversity-CIAT) or Andreas Wilkes. \ This document is a draft for internal review.